SKILL DOCUMENTATION

# CS Concept Tutor

Explain a concept, test it with MCQs, score and guide revision

Before vs. After comparison with screenshots


SUBMITTED BY

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# 1. About the Skill

**CS Concept Tutor** is a Claude skill built for Computer Science engineering (CSE) undergraduates preparing for college exams, placements and technical interviews. It teaches one concept at a time and then checks whether the student actually understood it with a short MCQ test. The goal is exam-ready and interview-ready understanding, not textbook completeness.

## Skill at a glance

| Item | Details |
|---|---|
| Skill name | `cs-concept-tutor` |
| Triggers when | A CSE student asks to explain, learn, understand, revise or be tested on any CS topic (OS, DBMS, DSA, networks, OOP, algorithms, compilers, architecture, system design basics). |
| Example prompts | "explain paging", "teach me normalization", "quiz me on deadlocks", "test me on trees", "what is a semaphore" |
| Output | Structured explanation, 5 MCQs, then score, explanations, revision advice and next steps |
| Code examples | Short C++ (under about 15 lines), only when the concept is about code; otherwise analogies or tables |

## How it works

After an initial level check, the flow runs in three phases over separate turns: explain and test, wait for answers, then evaluate.

| Phase | What the skill does |
|---|---|
| 0. Find the level | Uses the level the student states (for example, "first year" = Beginner, "placements" = Intermediate, "interview deep dive" = Advanced). If none is given, it asks once and waits instead of guessing. |
| 1. Explain + test | Gives a heading, a 1-2 sentence definition, an easy explanation, 3-6 key points and an example or analogy. Immediately follows with 5 scenario-based MCQs (A-D, one correct answer) and no answers revealed. |
| 2. Wait | Waits for answers in the format `1-B, 2-A, ...`. Doubts are answered without hinting at any test answer. |
| 3. Evaluate | Shows the score, a result table, explanations for wrong answers only, a diagnostic 'What to revise' note and next steps. |
| Follow-ups | Offers a retest (5 fresh MCQs on the weak area) or related topics, reusing the known level. |

## Difficulty levels

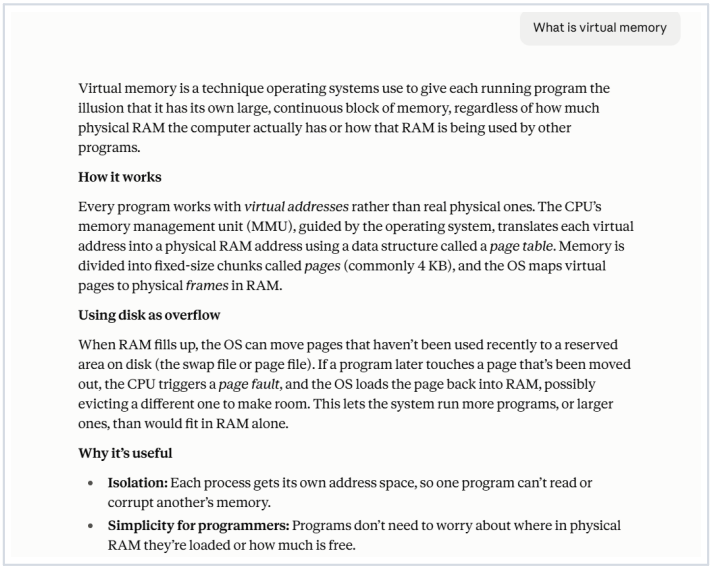
| Level | Explanation style | Question style |
|---|---|---|
| Beginner | Plain words, one analogy, every term defined | Direct understanding, simple scenarios |
| Intermediate | Standard exam/placement depth, correct terminology | Scenario-based, concept application |
| Advanced | Edge cases, complexity, trade-offs, comparisons | Tricky scenarios, design choices |

## Key design rules

- Questions test understanding through scenarios and 'what happens if' reasoning, not definition recall.
- Distractors reflect real misconceptions, and correct answers are spread across A-D.
- Answer keys are verified before sending so the score is trustworthy.
- Very broad topics (such as 'Operating Systems') are narrowed to one key subtopic instead of explaining everything.
- If the student wants to skip the test, the skill gives the answers with brief explanations and no scoring.
- Feedback is encouraging and direct, and explanations cover incorrect answers only.

# 2. Before: Without the Skill

Prompt used: **"What is virtual memory"**. Claude gives a general, one-size-fits-all explanation.



What is virtual memory

Virtual memory is a technique operating systems use to give each running program the illusion that it has its own large, continuous block of memory, regardless of how much physical RAM the computer actually has or how that RAM is being used by other programs.

**How it works**

Every program works with *virtual addresses* rather than real physical ones. The CPU's memory management unit (MMU), guided by the operating system, translates each virtual address into a physical RAM address using a data structure called a *page table*. Memory is divided into fixed-size chunks called *pages* (commonly 4 KB), and the OS maps virtual pages to physical *frames* in RAM.

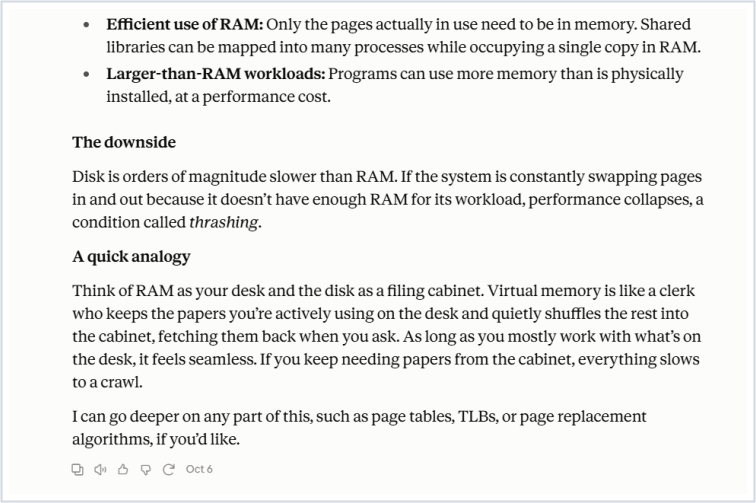
**Using disk as overflow**

When RAM fills up, the OS can move pages that haven't been used recently to a reserved area on disk (the swap file or page file). If a program later touches a page that's been moved out, the CPU triggers a *page fault*, and the OS loads the page back into RAM, possibly evicting a different one to make room. This lets the system run more programs, or larger ones, than would fit in RAM alone.

**Why it's useful**

- **Isolation:** Each process gets its own address space, so one program can't read or corrupt another's memory.
- **Simplicity for programmers:** Programs don't need to worry about where in physical RAM they're loaded or how much is free.

*Figure 1: Plain explanation with no level check (definition, how it works, disk as overflow).*



- **Efficient use of RAM:** Only the pages actually in use need to be in memory. Shared libraries can be mapped into many processes while occupying a single copy in RAM.
- **Larger-than-RAM workloads:** Programs can use more memory than is physically installed, at a performance cost.

**The downside**

Disk is orders of magnitude slower than RAM. If the system is constantly swapping pages in and out because it doesn't have enough RAM for its workload, performance collapses, a condition called *thrashing*.

**A quick analogy**

Think of RAM as your desk and the disk as a filing cabinet. Virtual memory is like a clerk who keeps the papers you're actively using on the desk and quietly shuffles the rest into the cabinet, fetching them back when you ask. As long as you mostly work with what's on the desk, it feels seamless. If you keep needing papers from the cabinet, everything slows to a crawl.

I can go deeper on any part of this, such as page tables, TLBs, or page replacement algorithms, if you'd like.

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*Figure 2: Remainder of the answer, ending with a generic offer to go deeper.*

## Observations

- The depth is not matched to the student, because no level is asked or used.
- No structured Key Points section and no exam-style definition to quote.
- No test, no score and no feedback, so the student cannot tell whether the concept stuck.
- Ends with an open-ended offer rather than a clear next step.

# 3. After: With the Skill

Prompt used: **"explain virtual memory"**. The skill is triggered and runs the full tutor flow.

## Step 1 and 2: Level check and beginner explanation

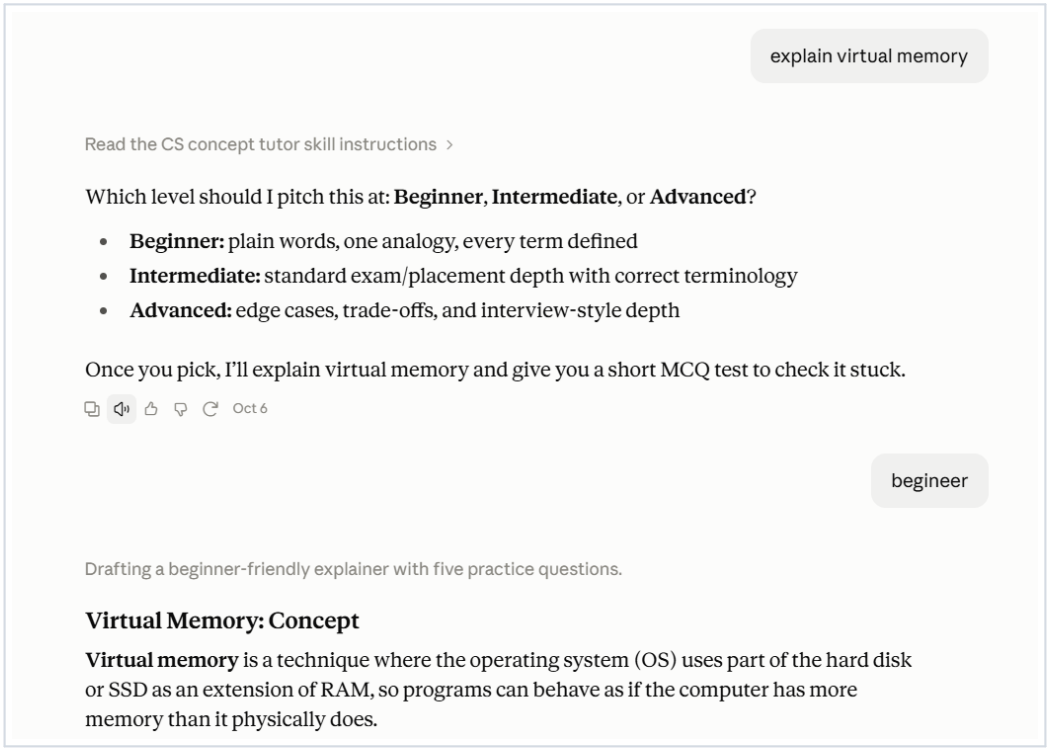



*Figure 3: The skill asks for the level once. The reply 'begineer' (typo) is understood as Beginner and the concept heading and definition follow.*

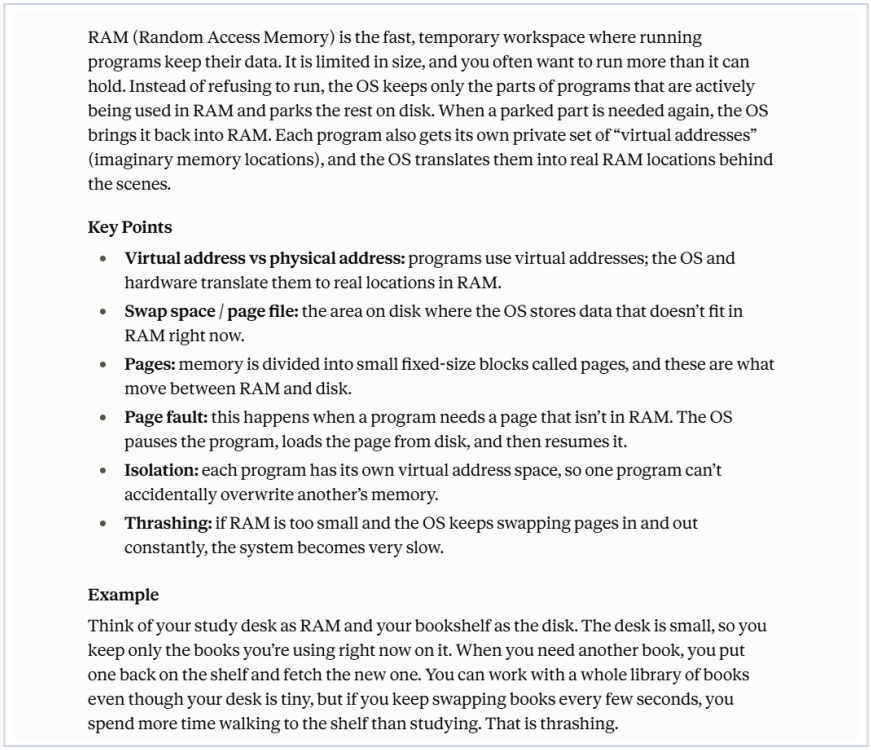



*Figure 4: Beginner-friendly explanation, six Key Points and a desk/bookshelf analogy.*

## Step 3: MCQ test

**Test Yourself**

1. A computer has 4 GB of RAM, but the programs you opened need about 6 GB in total. What does virtual memory allow the OS to do?
   A. Permanently delete the data that doesn't fit
   B. Refuse to open any program beyond 4 GB
   C. Keep some less-used data on disk and bring it back into RAM when needed
   D. Physically increase the size of the RAM chips
2. A running program tries to access data that is currently stored on disk and not in RAM. What happens?
   A. A page fault occurs, and the OS loads the needed page into RAM
   B. The program crashes immediately every time
   C. The CPU skips that instruction and continues
   D. The OS deletes the program from memory
3. Two programs both use the address 1000 in their own code, yet they don't overwrite each other's data. Why?
   A. The OS runs only one program at a time
   B. Address 1000 is reserved and unused by all programs
   C. The second program's data is always stored on disk
   D. Each program has its own virtual address space, and the OS maps the same virtual address to different physical locations
4. You open many large programs on a laptop with small RAM. The system becomes extremely slow and the disk activity light stays on constantly. What is the most likely cause?
   A. The CPU is too fast for the programs
   B. The OS is constantly swapping pages between RAM and disk because RAM is too small for the workload
   C. The page table has become too large to store on disk
   D. Virtual memory has made RAM larger, so everything runs faster

*Figure 5: 'Test Yourself' with scenario-based questions (Q1-Q4), four options each, no answers revealed.*

5. A user upgrades a computer from 4 GB to 16 GB of RAM while running the same programs as before. What is the most likely effect on virtual memory?
   A. Fewer page faults and less swapping, so the system usually runs faster
   B. Virtual memory stops working because there is too much RAM
   C. Programs can no longer use virtual addresses
   D. The page file must be made larger to match the new RAM

Reply with your answers in this format: `1-B, 2-A, 3-D, 4-C, 5-B`

1-B, 2-A, 3-D, 4-C, 5-B

**Your Score: 2/5**

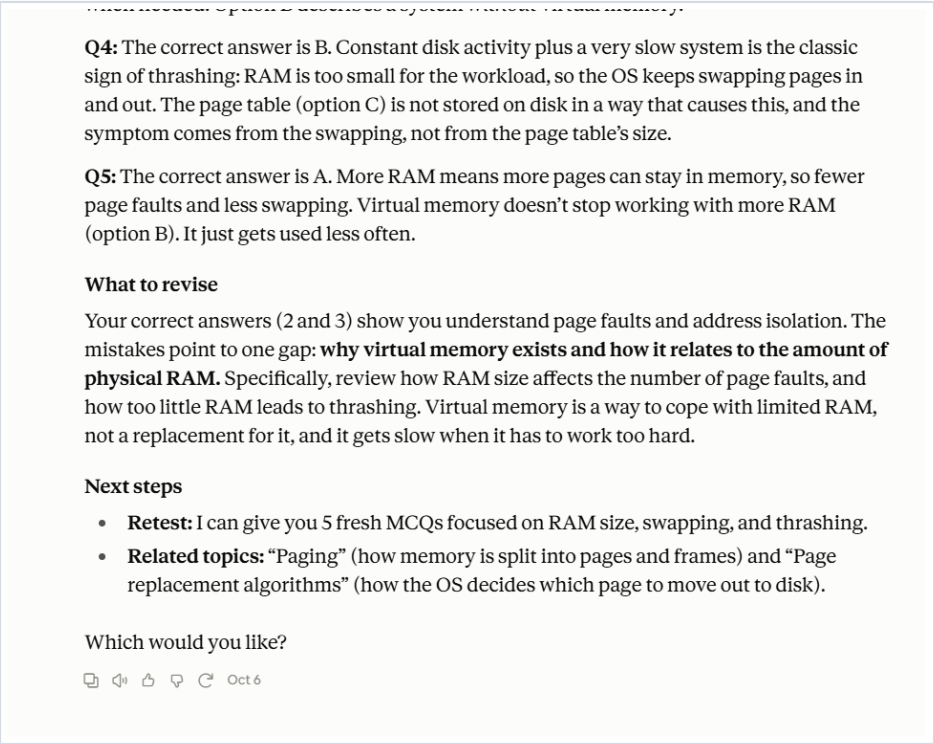
| # | Your Answer | Correct | Result |
|---|---|---|---|
| 1 | B | C | ✗ |
| 2 | A | A | ✓ |
| 3 | D | D | ✓ |
| 4 | C | B | ✗ |
| 5 | B | A | ✗ |

**Explanations for incorrect answers**

**Q1:** The correct answer is C. Virtual memory exists precisely so the OS doesn't have to refuse programs that exceed RAM. It keeps less-used pages on disk and brings them back when needed. Option B describes a system wit virtual memory.

*Figure 6: Question 5, the answer-format prompt, the student's reply (1-B, 2-A, 3-D, 4-C, 5-B) and the scored result table (2/5).*

## Step 4: Feedback and next steps



**Q4:** The correct answer is B. Constant disk activity plus a very slow system is the classic sign of thrashing: RAM is too small for the workload, so the OS keeps swapping pages in and out. The page table (option C) is not stored on disk in a way that causes this, and the symptom comes from the swapping, not from the page table's size.

**Q5:** The correct answer is A. More RAM means more pages can stay in memory, so fewer page faults and less swapping. Virtual memory doesn't stop working with more RAM (option B). It just gets used less often.

**What to revise**

Your correct answers (2 and 3) show you understand page faults and address isolation. The mistakes point to one gap: **why virtual memory exists and how it relates to the amount of physical RAM.** Specifically, review how RAM size affects the number of page faults, and how too little RAM leads to thrashing. Virtual memory is a way to cope with limited RAM, not a replacement for it, and it gets slow when it has to work too hard.

**Next steps**

- **Retest:** I can give you 5 fresh MCQs focused on RAM size, swapping, and thrashing.
- **Related topics:** "Paging" (how memory is split into pages and frames) and "Page replacement algorithms" (how the OS decides which page to move out to disk).

Which would you like?

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*Figure 7: Explanations for incorrect answers, a diagnostic 'What to revise' section, and the Retest / Related topics options.*

# 4. Before vs. After Summary

| Aspect | Before (no skill) | After (with skill) |
|---|---|---|
| Level handling | None; one generic depth | Asks level once and adapts content |
| Structure | Free-form answer | Definition, explanation, Key Points, Example |
| Active learning | Passive reading only | 5 scenario-based MCQs |
| Evaluation | None | Score table with correct answers marked |
| Feedback | None | Explanations for wrong answers only |
| Diagnosis | None | 'What to revise' names the underlying gap |
| Follow-up | Vague offer to go deeper | Retest or related topics, student chooses |

## Conclusion

The CS Concept Tutor skill turns a one-way answer into a complete learning loop: **level check, explain, test, score, diagnose, and continue**. In the screenshots, the student scored 2/5 and the skill pinpointed the exact gap (how RAM size relates to page faults and thrashing), then offered targeted retesting. That kind of feedback is not possible with a plain explanation.